

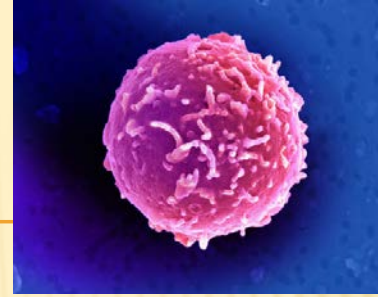
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CD34+ COLLECTION EFFICIENCY

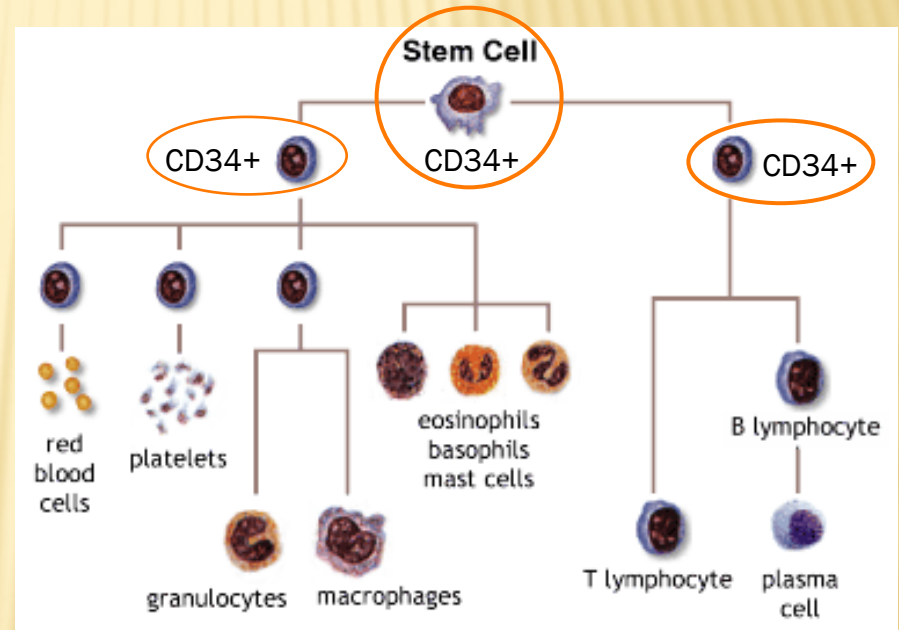


CD34+ cells and Collection Efficiency:
All you need to know about them

WHAT IS A CD34+ CELL?



- CD34 is a protein found in the membrane of very immature blood cells.
- Stem cells are among them, but other progenitor cells also express CD34.
- A CD34 positive cell is any cell that shows the CD34 protein in its membrane (stem cell or not).
- All CD34+ cells are MNC.
- MNC also includes mature cells: Lymphocytes and Monocytes.



HOW MANY CD34+ CELLS ARE REQUIRED FOR A TRANSPLANT?

- ✘ The FDA does not regulate CD34+ transplant doses, each Transplant Unit has its own. There is no such a thing as a universal Standard Dose.
- ✘ The most popular transplant dose is 5×10^6 per Kg for a single transplant, and twice that ($10 \times 10^6/\text{Kg}$) for a Tandem transplant (Multiple Myeloma) (*note: $10^6 = 1 \text{ Million}$*).
- ✘ Some oncologists accept any yield above $2 \times 10^6/\text{Kg}$.
- ✘ Transplant doses *below* $2 \times 10^6/\text{Kg}$ result in extended engraftment times, which put patient's lives at risk.

UNITS COMMONLY USED TO EXPRESS TARGET CELLS

Cell	Units	Examples	Definition
CD34+	Total CD34+	360 x 10e6	How many CD34+ in collection bag
	CD34+/Kg	4.6 x 10e6/Kg	M of CD34+ per Kg of recipient weight
	CD34%	0.05%,0.76%	% of CD34+ in WBC (pre-, product)
MNC	Total MNC	2.27 x 10e9	How many MNC in collection bag
	MNC%	15%,72%	% of MNC in WBC (pre-, product)

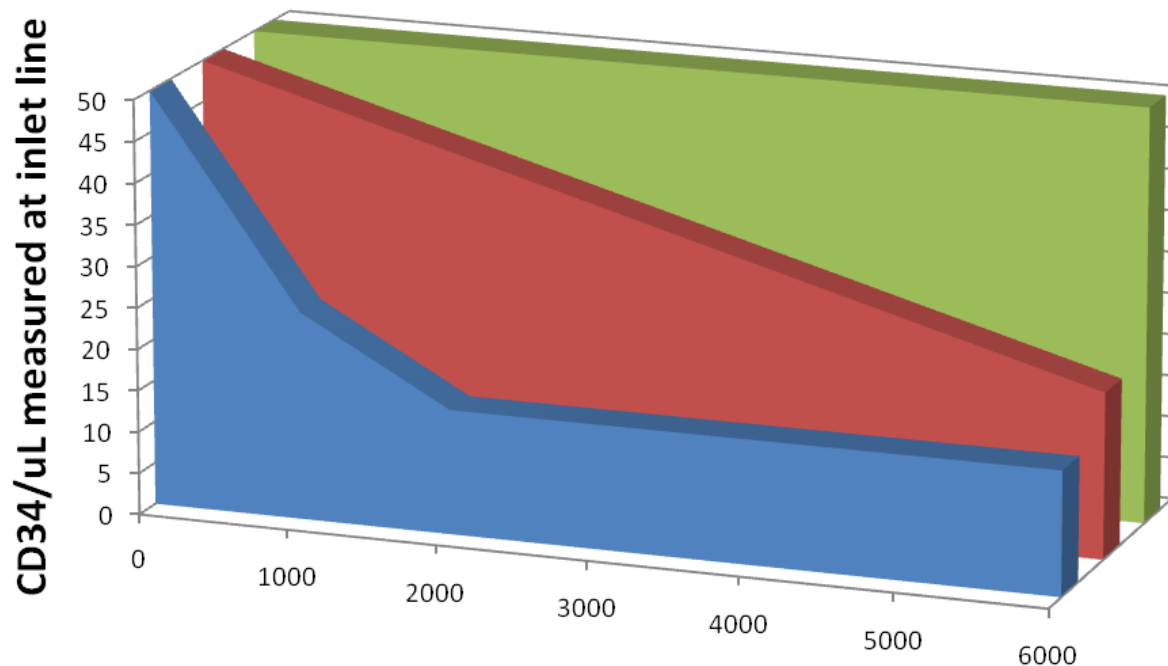
Note: $1 \times 10^6 = 1\text{E}+06 = 1 \times 10^6 = 1\text{M}$

COLLECTION EFFICIENCY - DEFINITION

- ✗ In plain English, Collection Efficiency (CE) could be defined as the percentage of cells going through the apheresis device that end up being collected in the collection bag.
- ✗ We can easily determine how many cells we collected at the end of the run, but determining how many of them went through the apheresis machine can be difficult.
- ✗ There are **three** models used to determine CE, and each one makes a different assumption to estimate how many cells went actually through the machine:

HOW MANY CD34+ CELLS WENT THROUGH THE DEVICE?

Model of CD34 change during apheresis



Inlet volume processed

■ Actual ■ CE1% ■ CE2%

	Collection Efficiency CD34
CE2	55.3%
CE1	62.7%
Actual*	85.1%

* Instantaneous

COLLECTION EFFICIENCY 2 (CE2)

Based on CD34+ **pre-count only**. The most popular in the US, because it doesn't require post CD34+:

$$CE2 = \frac{\text{Total CD34+ collected}}{\text{WB Volume} \times \text{pre-CD34+}}$$

- Since it assumes that CD34+ concentration in peripheral blood is constant throughout the procedure (not true), CE2 is not close to the Actual CE.
- In general, $CE2 < CE1$, but if there is CD34+ recruitment intra-procedure, the opposite happens.
- The more blood is processed, the lower the CE2 (the assumption about CD34+ cells going through the device gets worse)

COLLECTION EFFICIENCY 1 (CE1)

- Based on the **average** CD34+ count during the procedure (pre + post /2), therefore it requires a CD34+ *post* count. It is rarely used in the US because of the additional cost of the CD34+ post-count.

$$CE1 = \frac{\text{Total CD34+ collected}}{\text{WB Volume} \times \frac{\text{preCD34+} + \text{postCD34+}}{2}}$$

- CE1 is closer to the Actual CE as it assumes correctly that the concentration of CD34+ in peripheral blood is *not* constant.
- CE1 compensates for intra-procedure CD34+ mobilization.
- In general, CE1 > CE2.
- CE1 is pretty much immune to the influence of WB volume processed

WHAT DATA DO WE NEED TO CALCULATE CE2?

1. **Total CD34+** in collection bag
 1. Or donor Kg of body weight and CD34+/Kg.
 2. Or total WBC/ μ l in collection bag, % of CD34+ and product volume.
2. **WB liters** processed (excluding AC!)
3. **CD34+ pre-count** (donor)
4. A minimum of **20 procedures** is required for a reliable calculation of the median CE2 value.

Example

Total CD34+ in product x 10e6	360
pre-CD34+ / μ l	35
Processed WB (liters)	18.3
Collection Efficiency (CE2) %	56.2

WHY COLLECTION EFFICIENCY IS SO IMPORTANT?

- ✗ It is like a good mileage: *you can do more with less.*
- ✗ A good CE means that **less WB volume** will need to be processed to collect the required CD34+ dose for a transplant.
- ✗ Benefits of just processing the necessary WB volume:
 - + Shorter procedures if CD34+ pre-counts are high
 - + Lower Platelet Loss for the donor
 - + Less RBC, granulocytes in the product
 - + Smaller Product Volume
- ✗ CE is also a good way to monitor the quality of your collection procedures.

MOST COMMON QUESTIONS ABOUT CD34+ COLLECTION EFFICIENCY

- ✖ Why CE2 calculated by different centers using the same apheresis device is sometimes so different?
 - + Even if they use the same algorithm, *the way they take samples and count WBC and CD34+* makes a big impact in the final result.
 - + We can't compare directly CE2's calculated by different centers, BUT we can compare the CE2 of two different apheresis devices calculated in the same hospital.
- ✖ How can CE2 sometimes be so low (30's, 40's)? Is that number real?
 - + CE2 makes the incorrect assumption that CD34+ concentration in peripheral blood is the same throughout the procedure, so it underestimates the Actual CE.

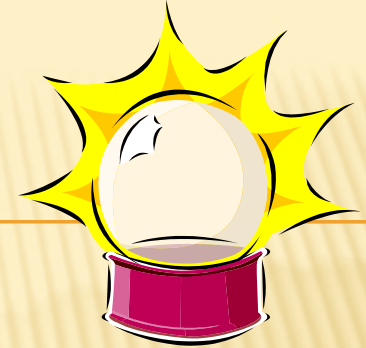
CD34+ COLLECTION EFFICIENCY (CE2) IN SOME US CENTERS

Center	Optia	Spectra
1	55%	39%
2	52.7%	N/A
3	54.2%	N/A
4	66%	60%
5	42%	35.1%
6	38.3%	27.4%
7	55.5%	38.3%
8	56%	43%
9	56.4%	30% (adjusted)
10	67.8%	58.0%
11	55%	46%

PREDICTION ALGORITHMS

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DEFINITION OF PREDICTION ALGORITHMS



- ✗ Prediction algorithms are mathematical calculations somehow based on the CE2 calculation, which is modified to obtain either the **CD34+ yield** or the **WB to process** to obtain a desired CD34+ yield:

$$\text{CD34+ Yield} = \text{CE2} \times \text{WB} \times \text{CD34+ pre-count}$$

or

$$\text{WB to process} = \frac{\text{Desired CD34+ yield}}{\text{CE2} \times \text{CD34+pre}}$$

THE ROLE OF PREDICTION ALGORITHMS IN PROCESS OPTIMIZATION



✘ Traditional process: One-size-fits-all

- + End point of MNC procedure **not** based on CD34+ pre-count or Collection Efficiency (CE) of apheresis device: Fixed time, fixed TBV's or fixed WB volume.
- + **Result: Outcome (CD34+ Yield) is unpredictable. Product volume and cross-cellular contamination can potentially be higher than desired.**

Product Volume	Product HCT	Product WBC	Product CD34+/μl	CD34+ /Kg
240	1.8	187.4	413	1.2
280	2.2	303.2	486	1.6
240	1.5	236.9	776	2.2
160	1.2	206.3	802	2.2
160	1.4	226.3	686	1.9
220	2.1	233.9	373	1.4
180	2.1	210.6	484	1.5
418	2.7	244.0	595	3.0
400	2.1	282.8	718	3.5
180	2.3	109.0	2,146	5.6
360	1.7	242.2	1,867	6.8
300	1.3	275.6	1,986	6.0
400	3.9	193.8	1,425	4.9
460	3.8	286.1	1,216	4.7
340	2.5	227.1	1,356	3.9
580	3.7	173.0	517	3.7
480	6.6	269.3	471	2.8
396	4.8	179.3	375	1.8
320	1.3	155.9	7814	29.4
340	1	184.7	8697	40.5

THE ROLE OF PREDICTION ALGORITHMS IN PROCESS OPTIMIZATION



× Optimized procedure: Using Prediction Algorithms

- + End point of MNC procedure (WB volume to process to achieve a desired yield) is based on **CD34+ pre-count** and **CE2** of apheresis device
- + Results:
 - × Predictable CD34+ yield
 - × Product volume and platelet loss under control

NMDP APPROACH (NMDP PROTOCOL V19.0, PAGE 34)

11.3.1. Overview of PBSC mobilization

The procedures for mobilization and collection of PBSC products under this protocol are dictated by the body weight of the proposed recipient. The body weight is not adjusted for lean-body mass.

The following table provides a guide to volume processed by weight of recipient. Volume processed refers to true whole blood volume, not including anticoagulant.

Table 3. Blood volume processed in relation to recipient weight.

Recipient Weight (kg)	Volume Processed (liters)	Procedure
≤ 35	12	Single 12-liter apheresis
36–45	15	Single 15-liter apheresis
46–55	18	Single 18-liter apheresis or two 12-liter procedures
56–65	22	Single 22-liter apheresis or two 12-liter procedures
> 65	24	Single 24-liter apheresis or two 12-liter procedures

Alternatively, centers may use an immediate pre-procedure blood CD34+ cell count to determine the blood volume to be processed. If such calculations are used, the negotiated CD34+ cell dose as specified in the final verification of PBSC request should be targeted as a minimum dose. If the negotiated CD34+ cell dose is collected in the first apheresis procedure, a second collection will not be performed.

The maximum total blood volume processed shall be 24 liters, whether collected over one or two day apheresis procedure(s).

PREDICTING THE WB VOLUME TO PROCESS

- ✗ It just requires
 - Median CD34+ CE2
 - Donor CD34+ pre-count
 - Desired CD34+ yield/Kg recipient
 - Kg recipient weight

1	Total CD34+ in product x 10e6	360
	pre-CD34+ / µl	35
	Processed WB (liters)	18.3
	Collection Efficiency (CE2) %	56.2
2	Desired CD34+ x 10e6 /Kg	5
	Kg recipient weight	75
	pre-CD34+ / µl	40
	CE2	56.2
	Liters of WB to process	16.7

- ✗ WB volume = $\frac{\text{Desired CD34+} \times 10^6 / \text{Kg} \times \text{Kg recipient}}{\text{CE2} \times \text{donor CD34+ pre}}$

Beware: 50% of the actual CD34+ yields will be above the target, but the other 50% will be below. Solution: use a conservative CE2.

TIME FOR QUESTIONS

